

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack Development Using MERN
Practical – I

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1.Environment Setup and JavaScript Basics

a. Install Node.js and Visual Studio Code, and verify installation.

```
PS C:\WINDOWS\system32> Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): _
```

```
● PS C:\MERN PRACTICAL> node -v
v24.17.0
● PS C:\MERN PRACTICAL> npm -v
11.13.0
```

b. Create a simple Node.js file and run it using the terminal.

Code:-

```
console.log("Hello, Welcome to 1st Practical of MERN");
console.log("Nandini");
```

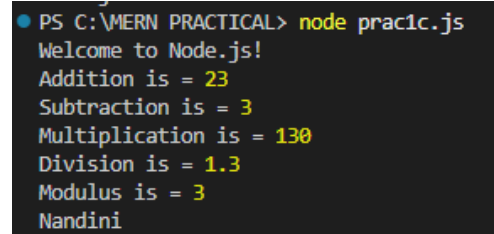
Output:-

```
● PS C:\MERN PRACTICAL> node prac1b.js
Hello, Welcome to 1st Practical of MERN
Nandini
```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

Code:-

```
console.log("Welcome to Node.js!");
let num1 = 13;
let num2 = 10;
console.log("Addition is =",num1+num2);
console.log("Subtraction is =",num1-num2);
console.log("Multiplication is =",num1*num2);
console.log("Division is =",num1/num2);
console.log("Modulus is =",num1%num2);
console.log("Nandini");
```

Output:-


```

PS C:\MERN PRACTICAL> node prac1c.js
Welcome to Node.js!
Addition is = 23
Subtraction is = 3
Multiplication is = 130
Division is = 1.3
Modulus is = 3
Nandini

```

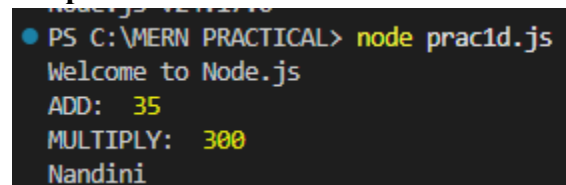
d. Create and run a simple program using variables and functions.

Code:-

```

console.log("Welcome to Node.js");
let a= 20;
let b= 15;
function add(a,b)
{
    return a+b;
}
function multiply(a,b)
{
    return a*b;
}
console.log("ADD: ",add(a,b));
console.log("MULTIPLY: ",multiply(a,b));
console.log("Nandini");

```

Output:-


```

PS C:\MERN PRACTICAL> node prac1d.js
Welcome to Node.js
ADD: 35
MULTIPLY: 300
Nandini

```

e. Demonstrate use of operators and control statements in JavaScript

Code:-

```

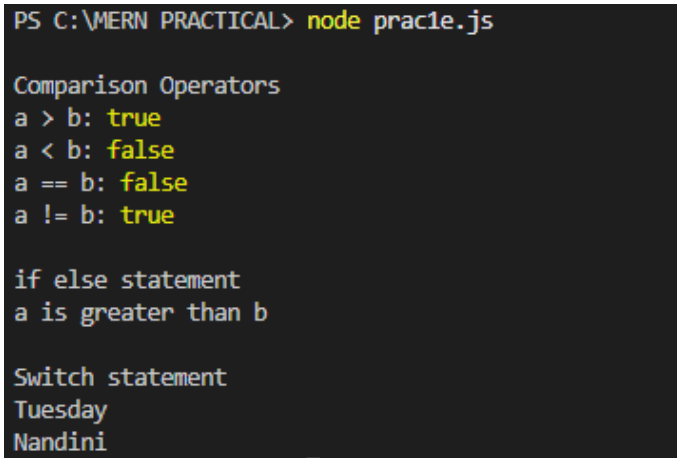
let a = 20;
let b = 10;
console.log("\nComparison Operators");
console.log("a > b:",a > b);
console.log("a < b:",a < b);
console.log("a == b:",a == b);
console.log("a != b:",a != b);
console.log("\nif else statement");
if(a>b) {
    console.log("a is greater than b");
}
else{

```

```

    console.log("b is greater than or equal to a");
  }
  //Switch Case
  console.log("\nSwitch statement");
  let day = 2;
  switch(day)
  {
    case 1:
      console.log("Monday");
      break;
    case 2:
      console.log("Tuesday");
      break;
    case 3:
      console.log("Wednesday");
      break;
    default:
      console.log("Invalid day");
  }
  console.log("Nandini");

```

Output:-


```

PS C:\MERN PRACTICAL> node prac1e.js

Comparison Operators
a > b: true
a < b: false
a == b: false
a != b: true

if else statement
a is greater than b

Switch statement
Tuesday
Nandini

```

f. Write programs using loops and template literals.**Code:-**

```

let num = 5;
console.log(`Multiplication Table of ${num}`);
for(let i=1; i <= 10; i++)
{
  console.log(`${num} x ${i} = ${num * i}`);
}
console.log("Nandini");

```

Output:-

```
PS C:\MERN PRACTICAL> node prac1f.js
Multiplication Table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
Nandini
```